

Nitikesh Vishal

Senior Backend Engineer | Python Developer | Data Engineer

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<https://techbrew.onrender.com/>

**Agile/Scrum Methodologies | Requirement Analysis | Offshore Coordination | Sprint Planning
Team Leadership | Cross-functional Collaboration**

Professional Summary

Senior Backend Engineer with 5.5+ years of experience building scalable backend platforms, microservices, and data engineering solutions across AI, Healthcare, ERP, SaaS, and E-commerce domains. Designed and delivered distributed systems processing millions of records, optimized APIs by up to 50%, and developed cloud-native applications on AWS and Azure. Expertise in Python, FastAPI, Django, PostgreSQL, Redis, and ETL architecture with a strong focus on performance optimization, system scalability, and enterprise-grade software engineering.

- Led requirement gathering sessions and collaborated directly with clients from the USA and Canada to define project scope and technical solutions.
- Managed onsite and offshore client engagements, ensuring effective communication, stakeholder alignment, and successful project delivery.
- Conducted daily scrum ceremonies, sprint planning, and progress tracking to maintain delivery timelines and Agile best practices.

Skills

Python



Flask Web
Framework



Redis



ETL



PySpark



AWS RDS



Django MVT



PostgreSQL



SQLAlchemy



Elastic Search



SciPy



AWS S3



Django REST



My SQL



Django ORM



Redis



Matplotlib



Microsoft Azure



FastAPI



Mongo DB



Data Pipelining



NumPy



Linux



Async/Await



GraphQL



Sqlite3



Luigi



Pandas



AWS EC2



Software
Development
Lifecycle(SDLC)



Agile 	Jira Project Tracking Tool 	JavaScript 	AJAX 	HTML 
Bootstrap 	CSS 	PHP 	Algorithms 	Data Structure 
Git 	GitHub 	Gitlab 	Bitbucket 	Docker 
Oops 	Jupyter Notebook 	Excel 	Team Leadership 	Problem Solving 

Career Experience

Turing Inc. – Off-Shore
Software Engineer L2 (Contract)

Bangalore (Remote)
October 2024 - March 2026

Worked on enterprise AI initiatives for global technology organizations, contributing to backend platform development, LLM training workflows, reinforcement learning systems, and scalable API infrastructure supporting next-generation AI applications.

Project: RL-Gym

Designed and developed scalable backend infrastructure for reinforcement learning ecosystems by building high-performance FastAPI services, PostgreSQL data models, and MCP-based integrations. Focused on creating maintainable microservice architectures, optimizing database performance, and enabling seamless interaction between AI workflows and backend systems.

- Architected FastAPI microservices supporting reinforcement learning workflows and serving thousands of API requests daily.
- Designed PostgreSQL schemas handling high-volume transactional and analytical datasets, reducing query execution time by nearly 40%.
- Improved API response latency by approximately 45% through asynchronous processing and query optimization.
- Developed secure authentication, authorization, and CRUD frameworks for enterprise-grade applications.
- Developed reusable backend modules, reducing development effort for new services by nearly 30%.

Project: Microsoft Copilot

Contributed to the development and evaluation of Large Language Model solutions through reinforcement learning, prompt engineering, response optimization, and human-feedback-driven training processes to improve model accuracy, safety, and alignment.

- Worked on RLHF pipelines to improve response quality and alignment.
- Collaborated with multidisciplinary teams to deliver production-ready AI workflows.
- Participated in knowledge-sharing initiatives around prompt engineering and model evaluation.

RnF Technologies – Onsite & Off-Shore
Software Engineer

Noida
January 2023 - May 2024

Led backend engineering efforts for SaaS and AI-powered platforms, building scalable APIs, data pipelines, cloud-native systems, and machine-learning integrations across enterprise applications.

Project: Scientific Data Integration Tool(SDI)

Skills involved: *FastAPI, PostgreSQL, Luigi, Async/Await, SQLAlchemy, Redis, Poppler, PyTorch, TensorFlow, Scikit-Learn, Numpy, Pandas, Microsoft Azure, Github, BitBucket, Linux, Jira, and VScode*

Designed and developed a data intelligence platform that transformed fragmented and unstructured datasets into centralized, searchable, and actionable business insights through automated data processing, ETL pipelines, AI integrations, and interactive visualization capabilities.

- Built end-to-end platform architecture from concept to production deployment serving enterprise data integration use cases.
- Engineered ETL pipelines using LUIGI, capable of processing millions of unstructured records from multiple data sources.
- Reduced data retrieval time by approximately 60% using PostgreSQL indexing and hierarchical data models.
- Implemented Redis caching mechanisms reducing repeated database hits by nearly 40%.
- Developed secure RBAC and JWT authentication supporting multi-tenant enterprise environments.

Project: JV Scan

Skills involved: *Python, Django REST, Django MVT, PostgreSQL, Librosa, AWS EC2, AWS S3, NGINX, Docker, AWS Cloud9, Gitlab, VsCode*

Developed an AI-powered healthcare platform capable of analyzing voice samples to generate health insights, reports, and predictive analytics aimed at improving accessibility to preliminary health assessments.

- Designed and developed AI-enabled healthcare APIs supporting real-time voice-processing workflows.
 - Integrated audio-processing and machine-learning workflows capable of processing thousands of audio samples.
 - Built scalable backend services achieving high availability and improved system reliability.
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**Hashstudioz Technologies Pvt. Ltd. - Onsite
Software Engineer**

**Noida
February 2021 - December 2022**

Developed enterprise-grade software solutions across education, healthcare, e-commerce, and media industries while working in distributed microservice environments serving large-scale business operations.

Project: Student Information System

Skills involved: *Django REST Api, Elastic Search, MySql, Docker, Jenkins, AWS EC2, AWS Cloud9, AWS CloudWatch, AWS S3, RDS, NGINX, GITLAB, Linux, Jira, VScode*

Contributed to the development of a large-scale ERP platform for Columbia University, supporting academic operations, student lifecycle management, and institutional workflows.

- Developed REST APIs supporting thousands of students and academic workflows across multiple institutional modules.
- Implemented Elasticsearch-based search systems indexing large datasets and significantly improving search efficiency.
- Optimized database queries and API performance, reducing response time by approximately 40%.
- Contributed to distributed microservices architecture supporting high availability and enterprise scalability.

Project: Bijnis

Skills involved: FASTAPI, Sanic, Async/Await, MongoDB, MySQL, Redis, PHP, Docker, Jenkins, AWS EC2, AWS S3, BitBucket, Linux, Jira, VScode

Contributed to the development of a B2B commerce platform connecting manufacturers, wholesalers, and retailers through scalable marketplace infrastructure and digital business workflows.

- Migrated legacy PHP services into FastAPI microservices, reducing API response latency by approximately 35%.
- Optimised MongoDB and MySQL operations, improving transaction throughput and database efficiency.
- Developed scalable seller and transaction management services supporting large-scale B2B commerce operations.
- Contributed to platform modernization initiatives resulting in improved maintainability and service reliability.

Project: Reuters Connect

Skills involved: GraphQL, Python, PostgreSQL, Scikit-Learn, TensorFlow, Docker, Jenkins, AWS EC2, AWS Cloud9, AWS S3, NGINX, GitLab, Linux, Jira Software Management, VScode

Built backend systems supporting global content distribution, media management, and information exchange workflows used by international publishers and media organizations.

- Developed GraphQL backend services supporting global media content distribution workflows.
 - Implemented secure encryption mechanisms using RSA-based key exchange protocols for sensitive transactions.
 - Designed backend services capable of supporting high-volume international publishing operations.
 - Built predictive analytics pipelines for scientific datasets and future trend forecasting.
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**Ideatebox - Onsite
Python Developer**

**Noida
September 2020 - February 2021**

Started professional software engineering journey by contributing to healthcare applications, backend systems, and enterprise web platforms while gaining expertise in Python development and software engineering best practices.

Project: ZenixSpace

Skills involved: Django MVT, Django REST, Python, MySQL, Docker, Jenkins, AWS EC2, AWS Cloud9, RDS, NGINX, Github, SSH, Linux, VS Code

Developed healthcare management software designed to streamline patient data management, content administration, authentication, and healthcare product workflows through a centralized digital platform.

- Developed backend modules using Django and Django REST Framework.
- Implemented secure authentication and authorization systems.
- Integrated Stripe payment processing.
- Built content management and administrative workflows.
- Assisted in deployment and infrastructure management.

Education

	AKTU University - B. Tech (CSE)	8.71 CGPA	Aug 2016 – July 2020
	Bachelor of Technology in Computer Science and Engineering with strong academic performance and a CGPA of 8.71. Developed a solid foundation in software engineering, algorithms, database systems, operating systems, computer networks, and object-oriented programming.		
	D.A.V Public School Higher Secondary	88%	(2013-2015)